Effects of Immigrants on Non-host Regions
Evidence from the Syrian Refugees in Turkey

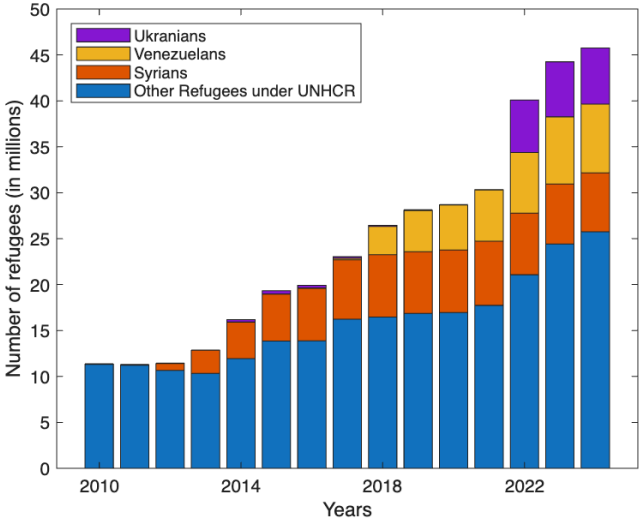
Ahmet Gulek

Stanford IPL, RFBerlin

June 19

Job Market Talk

Motivation: Rise in Refugee Crises Worldwide



Turkey Hosts 3.6 million Syrian Refugees

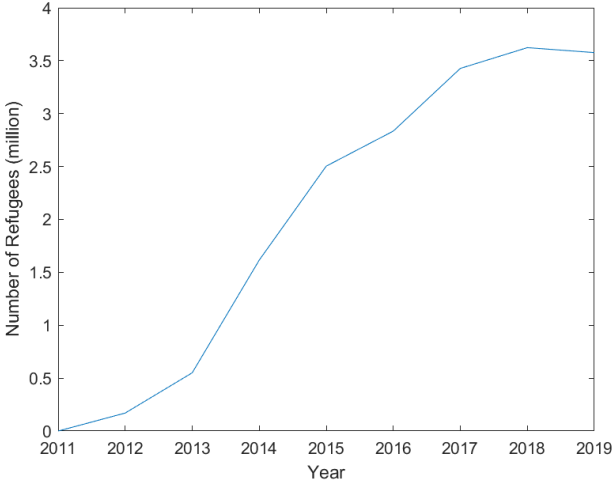


Figure: Timeseries of the number of Syrian refugees in Turkey

Shock is Large and Spatially Correlated

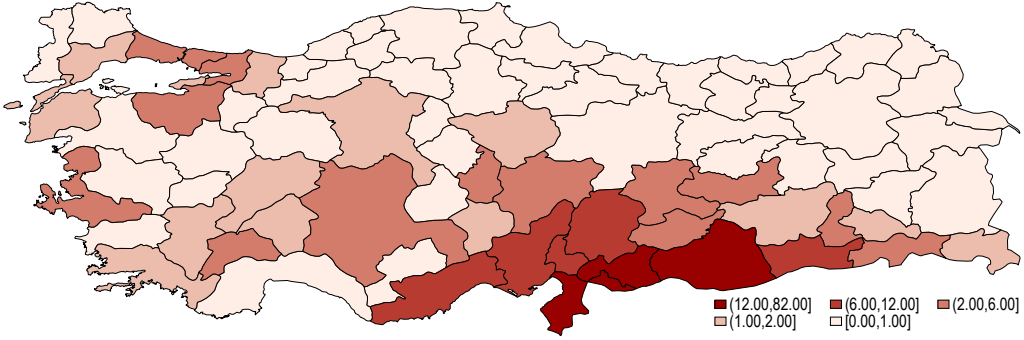


Figure: Number of Syrians per 100 natives in 2019

This Paper

Theory

Empirics

Counterfactuals

This Paper

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- Immigrants' effects propagate both forward and backward
- Effects depend on network and two elasticities
  - EoS b/w labor and intermediates ; EoS b/w intermediates

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  - In 71/81 regions, 1% increase in labor supply $\Rightarrow \Delta w_{host} \approx -1\%$, $\Delta w_{rest} \leq 0.01\%$
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- Larger welfare gains from high-skill immigration & immigration to central nodes

Contributions to the Literature

- **Effects of immigration on labor markets**
  - Card (1990); Altonji and Card (1991); Borjas et al. (1997); Card (2001); Borjas (2003); Card (2009); Ottaviano and Peri (2012); Dustmann et al. (2016); Borjas (2017); Peri and Yasenov (2019); Beerli et al. (2021); Doran et al. (2022); **Gulek (2024); Bahar, Di Tella, Gulek (2024)**
  - **Methodological debate:** Card (1990, 2009) assume local effects vs. Borjas (2003) argues spillovers
  - **Contribution:** immigrants affect labor markets in non-host regions via input-output networks
  - **Implication:** economic centrality determines when SUTVA fails; explains divergent findings

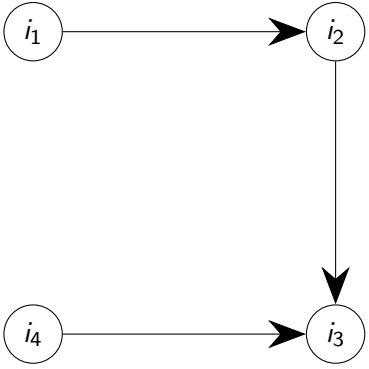
- **Trade and immigration interaction**
  - Dustmann and Glitz (2015); Burstein et al. (2020); Caliendo et al. (2021); Brinatti (2024); Akgündüz et al. (2024)
  - **Contribution:** production network structure shapes immigration effects

- **Propagation of shocks through production networks**
  - Acemoglu et al. (2012); Hornbeck (2012); Autor et al. (2013a,b); Hornbeck and Naidu (2014); Diamond (2016); Acemoglu et al. (2016a,b); Barrot and Sauvagnat (2016); Baqaee and Farhi (2019); Boehm et al. (2019); Notowidigdo (2020); Carvalho et al. (2021)
  - **Contribution:** extend network propagation framework to labor supply shocks
  - **Contribution:** welfare gains from immigration to central nodes (i.e., large cities)

Road Map

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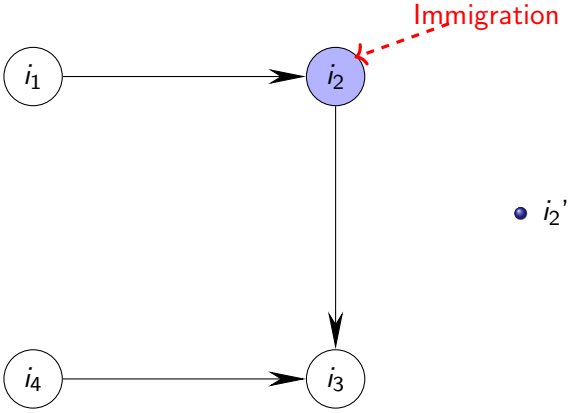
Basic Input-Output Network



Description:

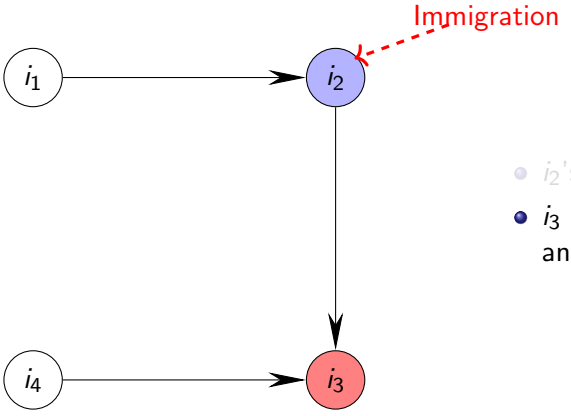
- i_1 supplies i_2.
- i_2 supplies i_3.
- i_4 also supplies i_3.

Step 0: Immigration Lowers Wages and Prices in Host Region



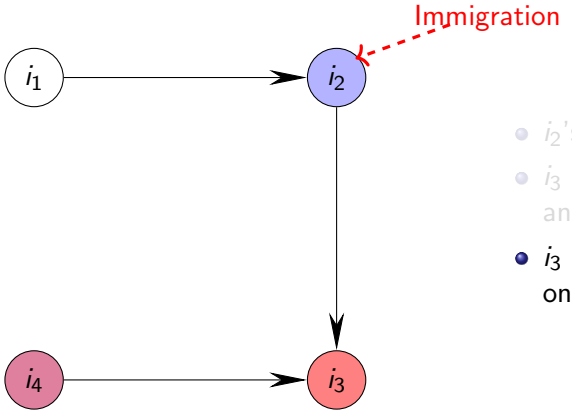
- i_2's labor costs and prices go down

G.E. Force 1: Cost Shock Propagates Forward



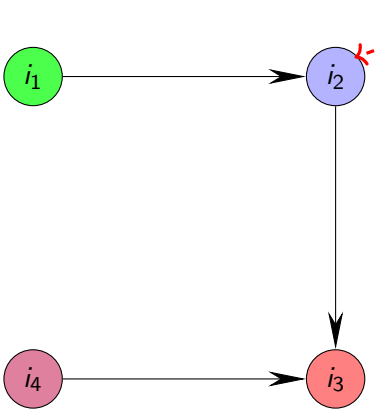
- i_2's labor costs and prices go down
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G.E. Force 2: Demand Spillovers from Substitution among Intermediates



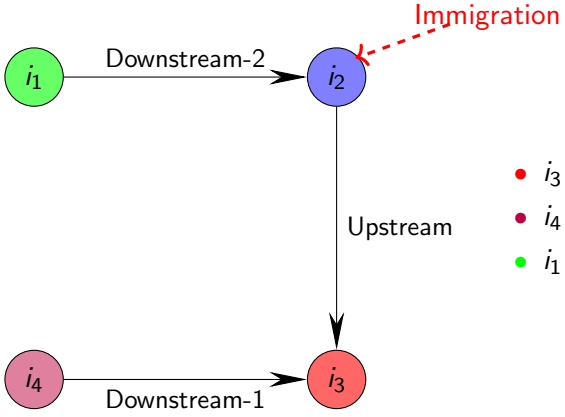
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- i_3 reallocates its spending between i_2 and i_4 based on substitutability among intermediates

G.E. Force 3: Demand Spillovers from Substitution



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- i_3 reallocates its spending between i_2 and i_4 based on substitutability among intermediates
- Two forces impact i_1
  1. i_2's input choice between labor and i_1's goods
  2. i_3's reallocation between i_2 and i_4

Naming Trade Spillovers



- i_3 is Upstream exposed
- i_4 is Downstream-1 exposed
- i_1 is Downstream-2 exposed

Model Overview

Labor and Intermediate Good Demand

$$\min_{\{x_{ij}\}_{j=1}^{n}, L_i} \sum_{j=1}^{n} p_j x_{ij} + w_{r_i} L_i \qquad s.t.$$

$$A_i(\eta_i m_i^{\frac{\sigma_u - 1}{\sigma_u}} + (1-\eta_i) L_i^{\frac{\sigma_u - 1}{\sigma_u}})^{\frac{\sigma_u}{\sigma_u - 1}} \geq y_i$$

$$m_i = \left(\sum_{j=1}^{n} \alpha_{ij} x_{ij}^{\frac{\sigma_l - 1}{\sigma_l}} \right)^{\frac{\sigma_l}{\sigma_l - 1}}$$

Labor Supply

- Perfectly inelastic, immobile across regions, perfectly mobile across firms

Final Demand

- Final goods consumption and firm ownership are local

$$\max_{\{x_{0,i}\}} \Pi_{i \in r} x_{0,i}^{\beta_i} \quad s.t. \quad \sum_{i \in r} p_i x_{0,i} = w_r L_r + \sum_{i \in r} \pi_i$$

▸ Formal model & Theorem 1

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2. Effects of (U,D1,D2) on firms' labor share and sales map to two EoS parameters
3. Use EoS parameters and network to calculate immigration's effects on wages and welfare

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Three Identification Challenges

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Employ Shift-share IV to Address Endogeneity

1. **Endogeneity**: Immigrants choose where to locate based on economic conditions
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$$Z_{r,t} = \underbrace{\sum_{g=1}^{13} \frac{1}{d_{r,g}} \lambda_g}_{\text{Share}} \times \underbrace{N_t^{Syrian}}_{\text{Shift}}$$

- $d_{r,g}$: travel distance between Turkish region r and Syrian governorate g
- λ_g: Relative distance to Turkey $\times$ Population share
- N_t^{Syrian}: Number of Syrians in Turkey in year t

▸ Heatmap ▸ First-stage

Employ Synthetic Controls to Address Unobserved Confounders

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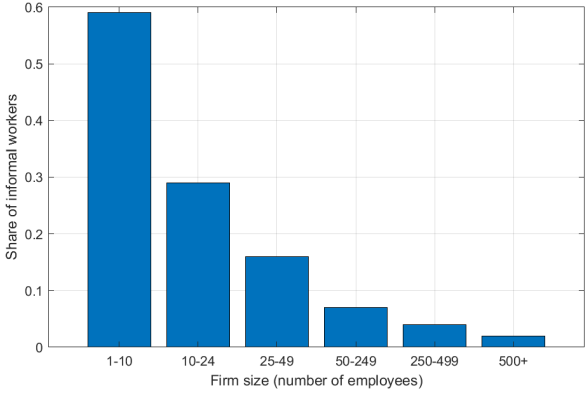
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Resulting Synthetic IV estimator is consistent (Gulek and Vives-i Bastida, 2024)

Informality Decreases by Firm Size

4. **Informality**: Small firms misreport costs and sales
   - Solution: Focus on large firms



Upstream Exposure Increases Firm Size

$$\widetilde{log(y_{it})} = \sum_{t' \neq 2011} \beta_{1,t'} \widetilde{U_i^Z} \mathbb{1}\{t = t'\} + f_i + f_t + \nu_{it}$$

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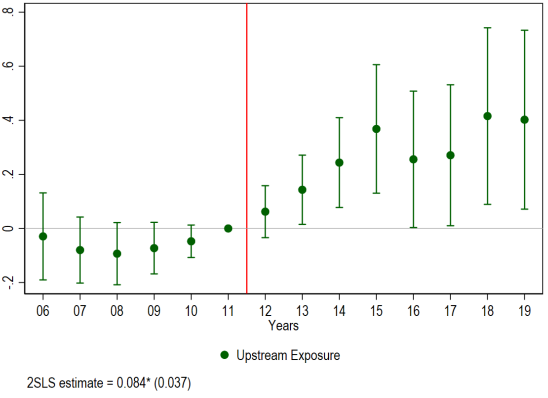


Figure: Upstream Exposure Effects on Size

Upstream Exposure Increases Payroll

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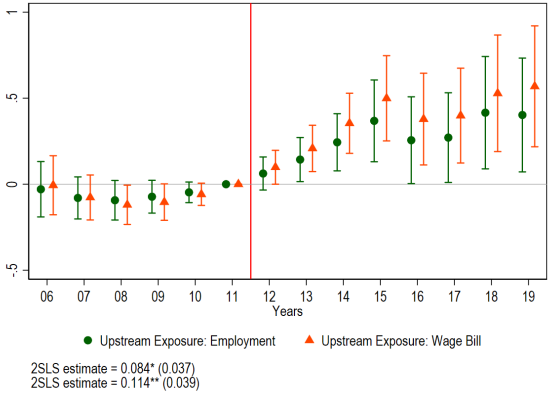


Figure: Upstream Exposure Effects on Size and Payroll

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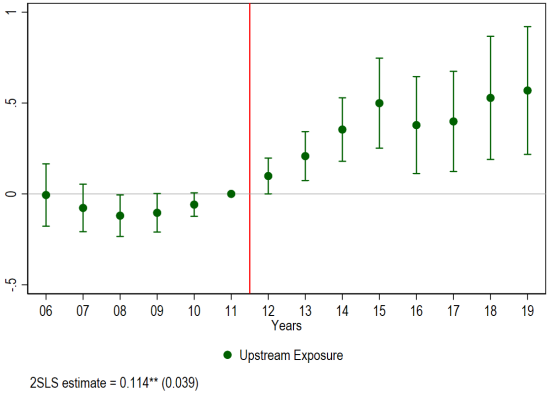
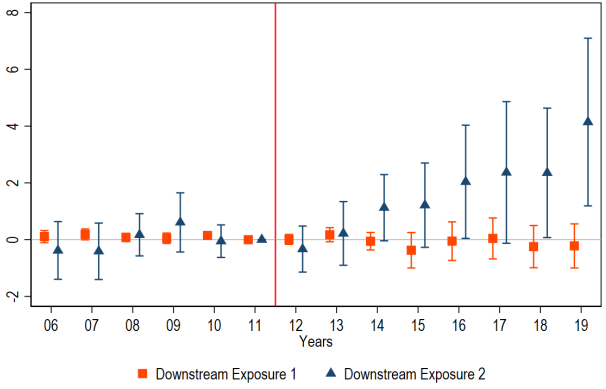


Figure: Upstream Exposure Effects on Labor Share

Effects of D1 and D2 exposures on sales

$$\widetilde{log(y_{it}^S)} = \sum_{t' \neq 2011} \left(\beta_{t'}^{D1} \widetilde{D1_i^Z} + \beta_{t'}^{D2} \widetilde{D2_i^Z} \right) \mathbb{1}\{t = t'\} + f_i^S + f_t^S + \nu_{it}^S$$



D1: 2SLS estimate = -0.031 (0.100)
D2: 2SLS estimate = 1.045* (0.465)

Elasticity Estimates

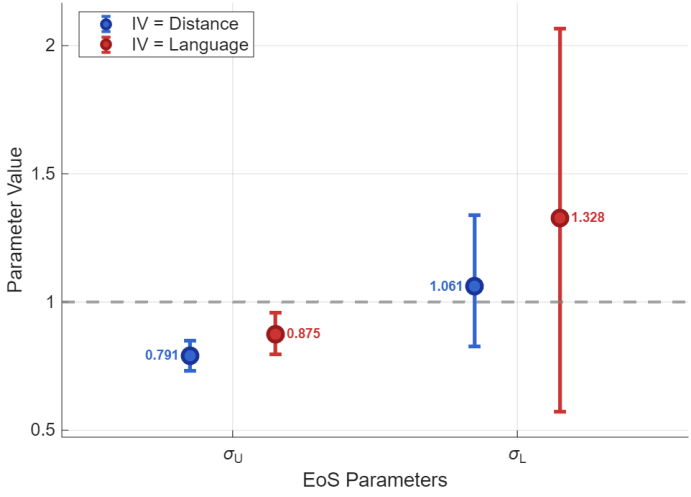


Figure: Elasticity of Substitution Estimates

Robustness Checks

- Language-based instrument: ▸ Alt IV
- Industry Heterogeneity: ▸ Nace-2
- SIV Validation:
  - IV alone not enough: ▸ IV-RF
  - Over-fitting: no pre-trends in **unmatched** outcomes
  - Under-fitting: no pre-trends in **matched** outcomes
  - Back-testing: ▸ Alternative Training
- Firm Size sensitivity: results on cost-based measures robust ▸ Size
- Sensitivity to calibrated ϵ^D: ▸ ϵ^D

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- **Take-away: mostly negligible spillovers**

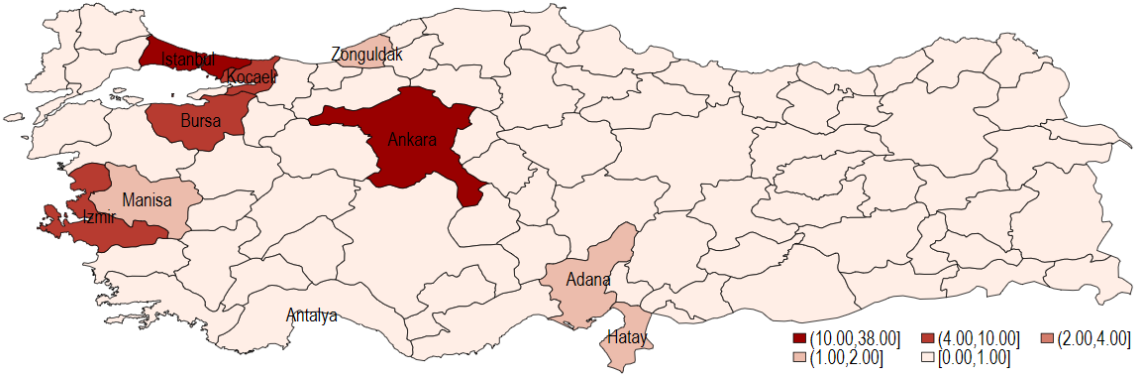
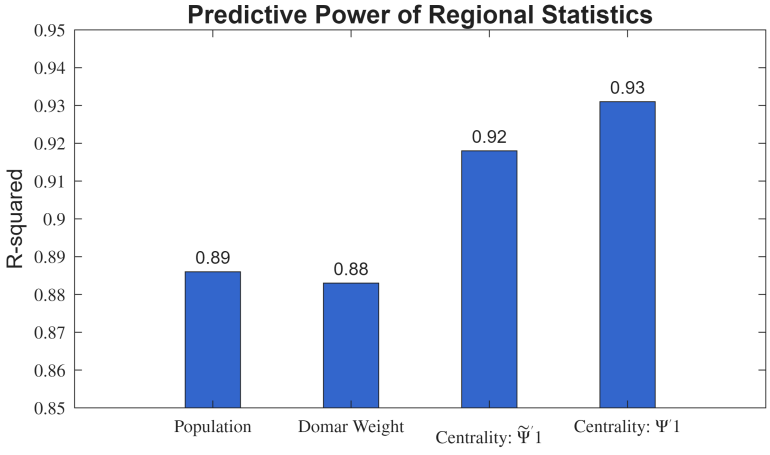


Figure: Heatmap of Non-host Region Wage Effects

Bonacich Centrality predicts spillovers



▸ Regression Table

Counterfactual 2: Larger Welfare Gains from Central Nodes

- Policy question: where should immigrants/refugees be located?
- Fix levels: 10,000 immigrants, compare welfare $d \ln Y = \frac{1}{81} \sum_{r=1}^{81} d \ln Y_r$

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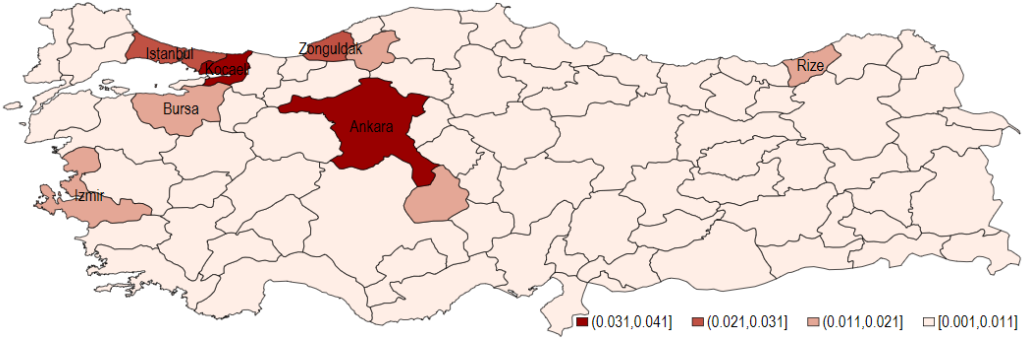


Figure: Heatmap of Total Welfare Effects of Immigration across Host Regions

Counterfactual 3: Higher the Skill, Larger the Spillovers

- Extended model to incorporate skill heterogeneity
- 10,000 low-skill vs 10,000 high-skill immigrants

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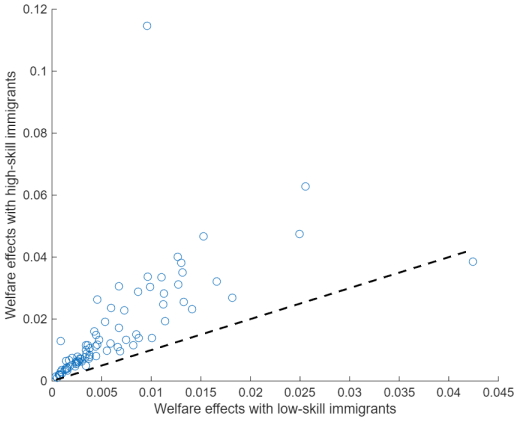
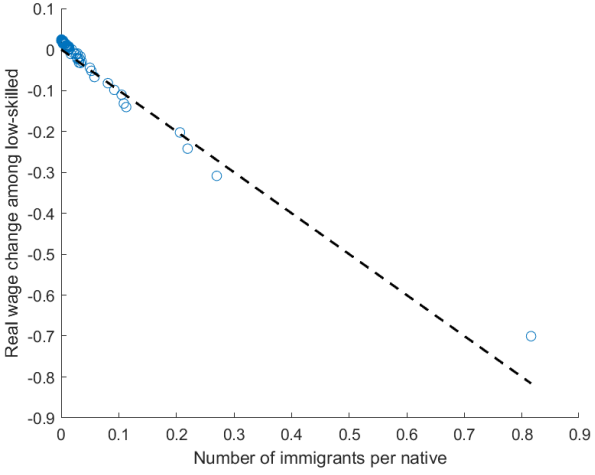


Figure: Comparison of welfare effects across low-skill and high-skill immigration

Model-based Factual: Aggregate Effect of the Syrian Shock

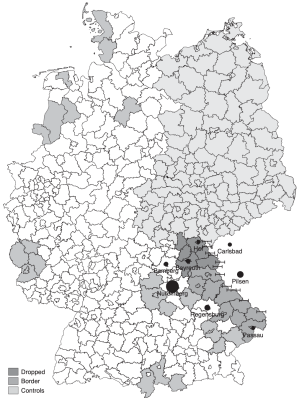


- Simulate the actual spatial distribution of Syrians
- Points lie on the -45° line: **GE** $\approx$ **PE**
- Syrians settled in non-central regions $\Rightarrow$ negligible aggregate spillovers

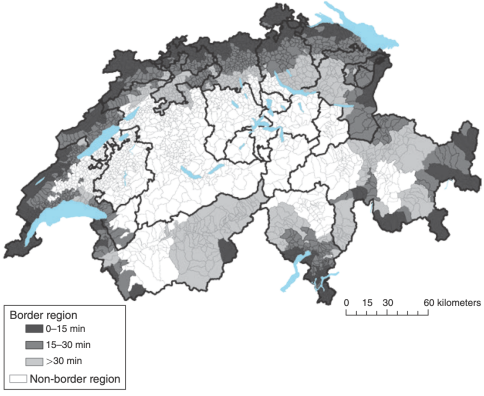
Road Map

1. Introduction
2. Intuition
3. Empirical Analysis
4. Counterfactuals
5. Implications
6. Conclusion

Implication 1: Resolving Puzzles — Dustmann et al. vs. Beerli et al.



Dustmann et al. (2017): **low-skill** → small German border regions (non-central) ⇒ no spillovers



Beerli et al. (2021): **high-skill** → Swiss border incl. Basel (central hub) ⇒ large spillovers

Same border-region design, opposite results. Inter-host spillovers can explain why

Implication 2: A Test of Productivity vs. Labor-Supply Shocks

- Labor-supply shock $\Rightarrow$ propagates **forward and backward** (this paper)
- Productivity shock $\Rightarrow$ **forward only** (Acemoglu et al., 2012)

$\Rightarrow$ the direction of I/O propagation is a test of the mechanism

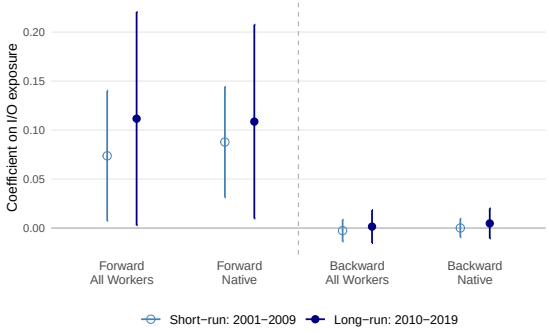
▸ STEM vs. non-STEM

Implication 2: A Test of Productivity vs. Labor-Supply Shocks

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H-1B (Abramitzky, Boustan, Gulek, Hainmueller, 2026): forward-only $\Rightarrow$ productivity, not LS, shock



H-1B income effects: forward > 0, backward ≈ 0

(preliminary)

▸ STEM vs. non-STEM

Implication 3: Production Networks — A Unifying Lens for Macro-Labor

- Cost/wage shocks propagate along the supply network
  - This paper: **immigration** (labor-supply shock)
  - Next paper (JMP): **minimum-wage** shocks — microfounds worker reallocation from more- to less-exposed firms

Production networks may be a unifying lens for two central literatures in macro-labor: immigration and the minimum wage.

Conclusion

- **This paper:**
  - Formalizes how immigration-induced wage changes propagate through production networks
  - Isolates relevant economic forces and tests them empirically using novel methods
  - Quantifies immigration's total effects on wages and welfare across regions

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  - Formalizes how immigration-induced wage changes propagate through production networks
  - Isolates relevant economic forces and tests them empirically using novel methods
  - Quantifies immigration's total effects on wages and welfare across regions

- **Key implications:**
  - Resolves puzzles in immigration literature: economic centrality explains divergent findings
  - Guides practitioners: immigration to large cities violates SUTVA; smaller regions allow reliable spatial DiD designs
  - Informs policy: immigrants improve welfare when they settle in central regions
    - Future work: trade-off between centrality and congestion externalities

Heatmap of Distance Exposure

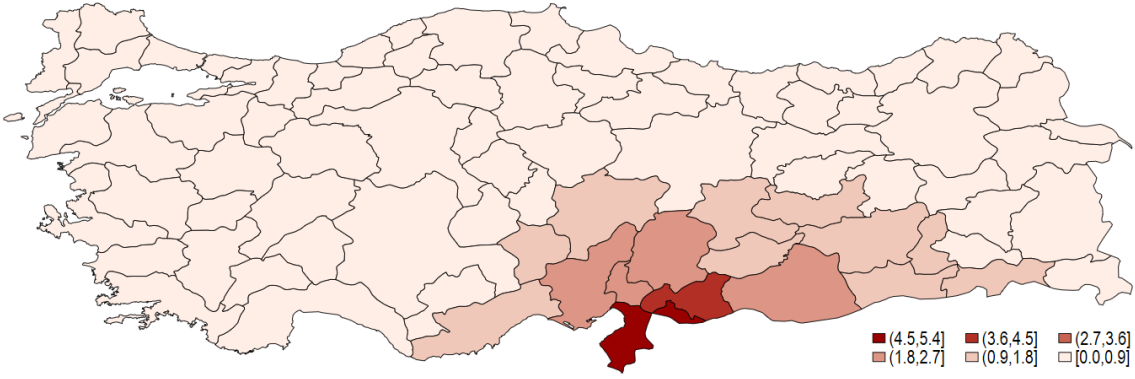
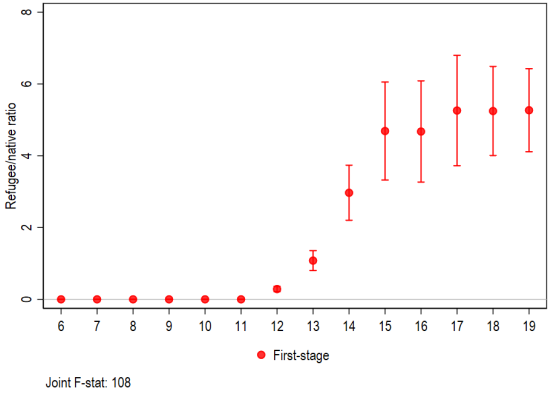


Figure: Distance IV heatmap

Return

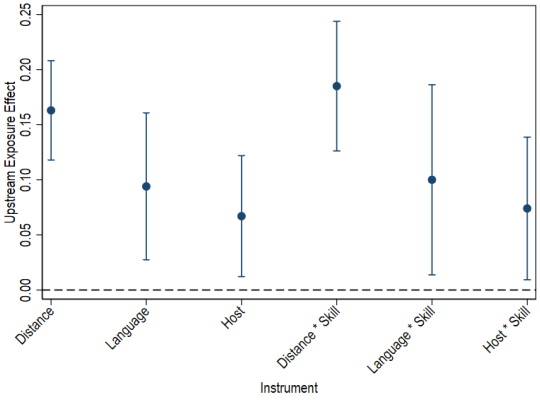
First-stage

$$\delta_{rt} = \sum_{t' \neq 2011} \beta_{t'} \text{Share}_r \mathbb{1}\{t = t'\} + f_r + f_t + \epsilon_{rt}$$

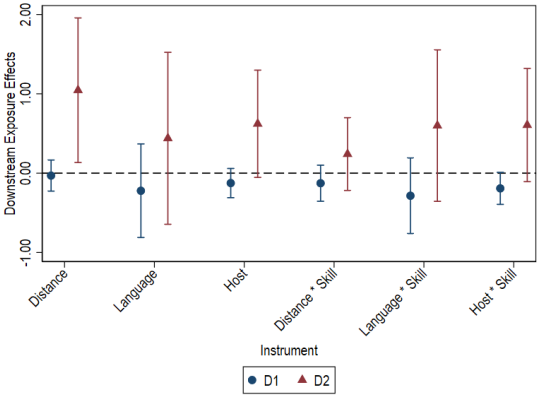


Return

Alternative IVs



(a) Labor share

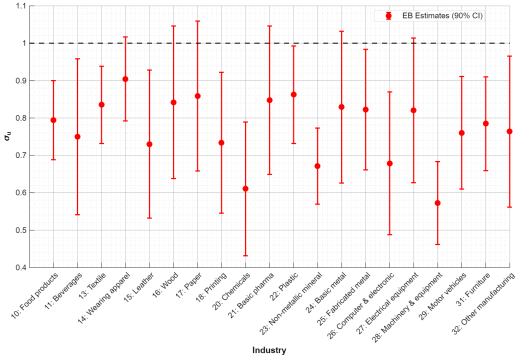


(b) Sales

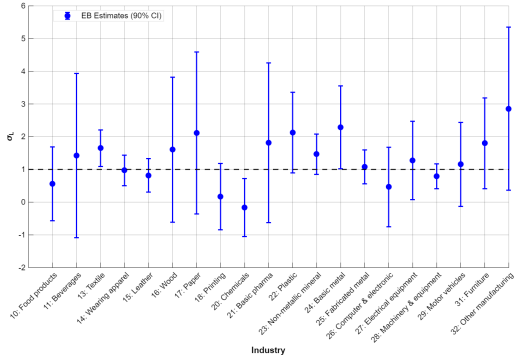
Figure: Effect of Trade Exposures on Labor Demand and Sales under Alternative Instruments

Return

Industry Heterogeneity (1/2)



(a) EoS between labor and intermediates



(b) EoS between intermediates

Figure: Heterogeneity of EoS estimates across Manufacturing industries

▸ More ◂ Return

Industry Heterogeneity (2/2)

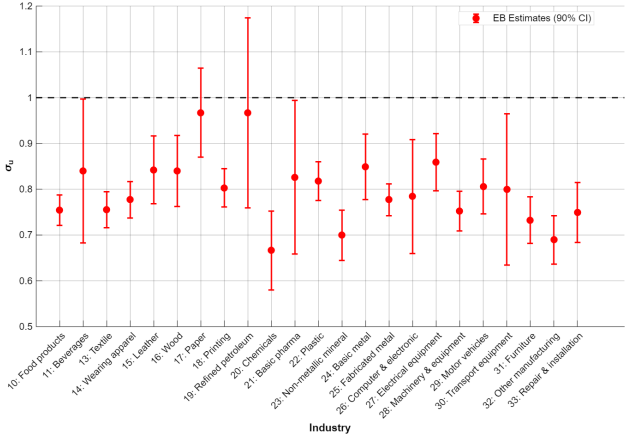
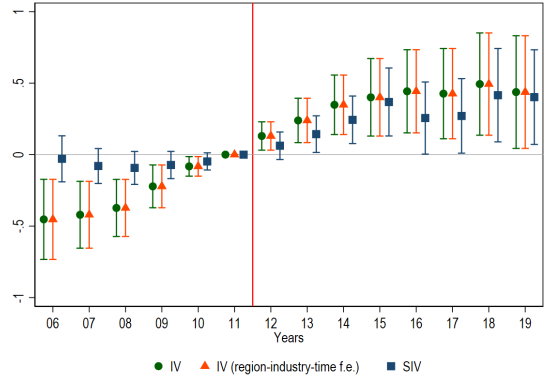


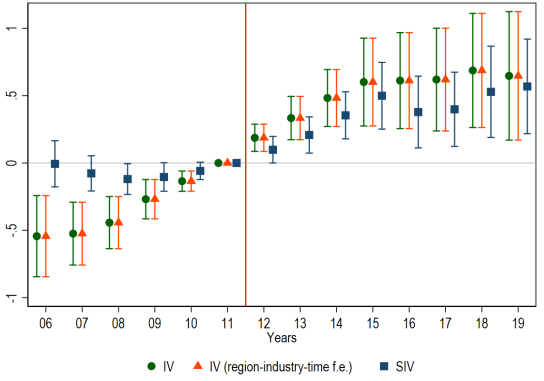
Figure: EoS between labor and intermediates (all manufacturing firms)

Return

IV vs SIV: Cost measures



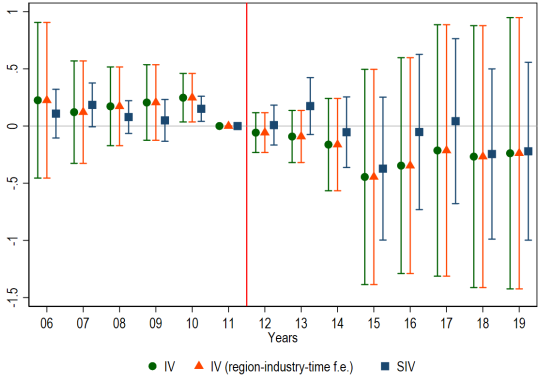
(a) Number of employees



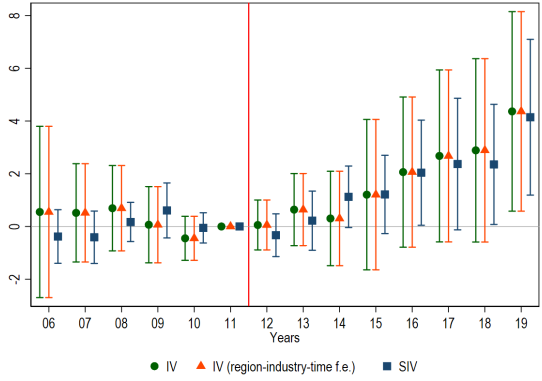
(b) Payroll

Sales Return

IV vs SIV: Sales measures



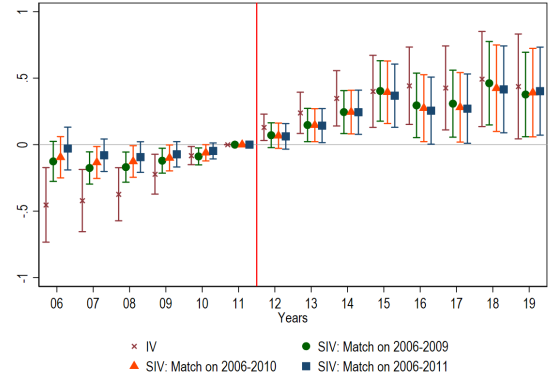
(a) Sales: Downstream-1



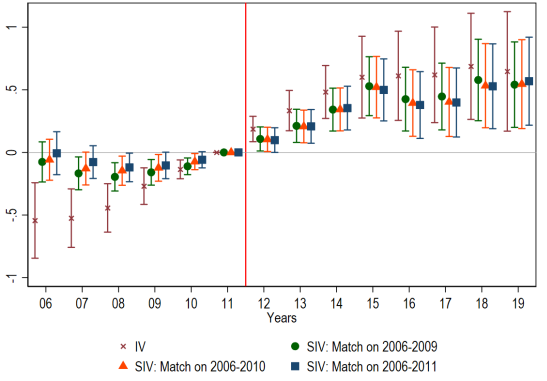
(b) Sales: Downstream-2

Return

Alternative Training Periods: Cost measures



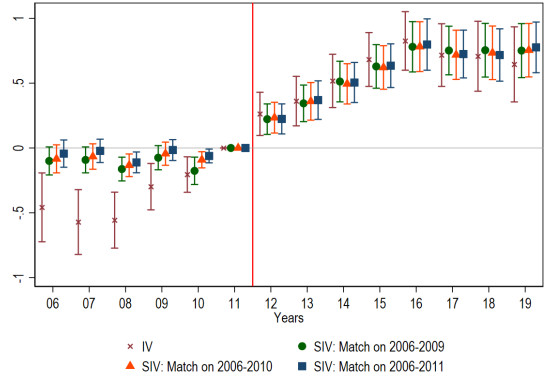
(a) Number of employees



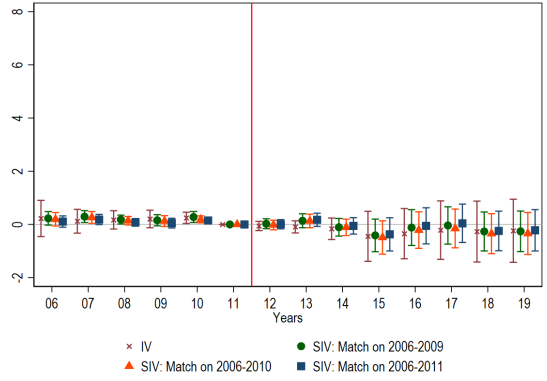
(b) Payroll

▸ More ◂ Return

Alternative Training Periods: Cost measures



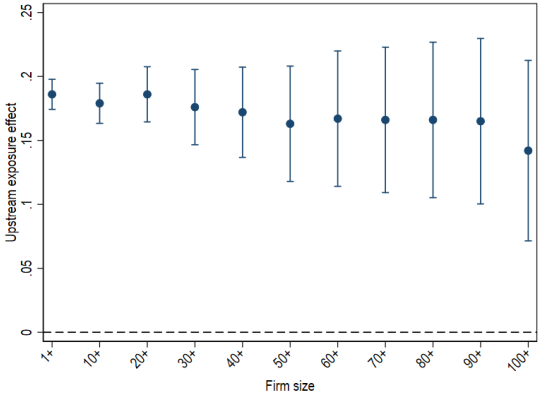
(a) Labor share



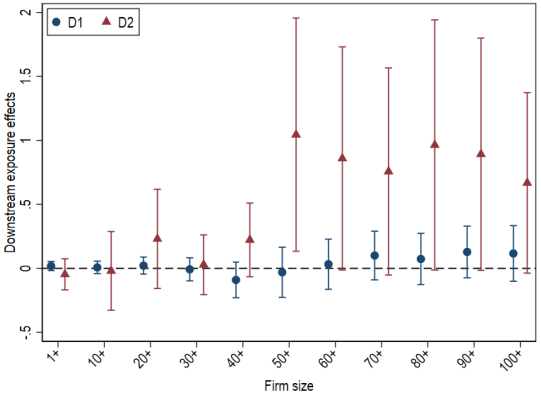
(b) Sales: D1-exposure

Return

Robustness to Size Selection



(a) Upstream Exposure Effects on Labor share

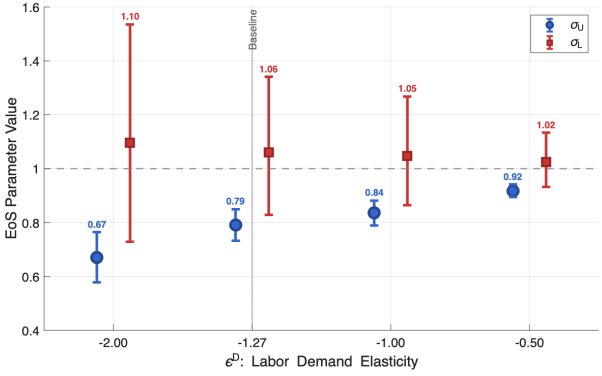


(b) Downstream Exposure Effects on Sales

Figure: Estimated Effects of Trade Exposures Across Different Firm Sizes

Return

Structural Elasticities: Sensitivity to Calibrated ϵ^D



- Qualitative conclusions hold for any $\epsilon^D < 0$: $\sigma_u < 1$ (complements), $\sigma_l \approx 1$ (Cobb-Douglas)

Return

Bonacich Centrality predicts spillovers

Table: Provincial Attributes and Spillovers from a 1% increase in Labor Supply

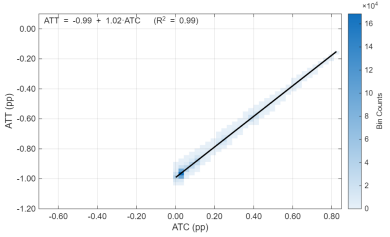
	(1) $\Delta W_{Non-host}$	(2) $\Delta W_{Non-host}$	(3) $\Delta W_{Non-host}$	(4) $\Delta W_{Non-host}$	(5) $\Delta W_{Non-host}$
Population	0.046***				0.0070
	(0.003)				(0.005)
Domar weight		0.046***			-0.079**
		(0.004)			(0.033)
Cost-Based Centrality: $\tilde{\Psi}\mathbf{1}$			0.047***		-0.024
			(0.004)		(0.039)
Sales-Based Centrality: $\Psi\mathbf{1}$				0.047***	0.14**
				(0.004)	(0.064)
N	81	81	81	81	81
R-sq	0.886	0.883	0.918	0.931	0.967

Note: All explanatory variables are standardized to have mean zero and standard deviation of 1. Robust standard errors are used. * 0.1 ** 0.05 *** 0.01

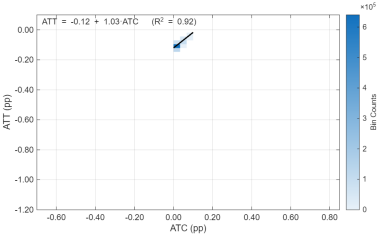
Return

Appendix: ATT vs. ATC Across Technologies and Outcomes

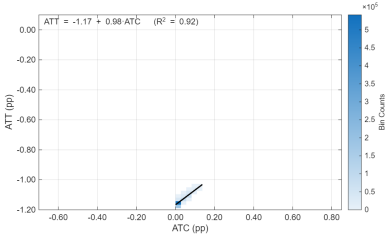
1M simulations; each assigns 10 of 81 regions a 1% labor-supply shock. Under SUTVA, ATC = 0.



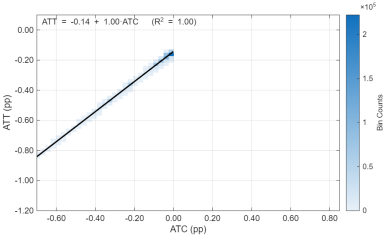
Real wages: σ_u=0.79, σ_l=1.06



Real wages: σ_u=10, σ_l=1 (substitutes)



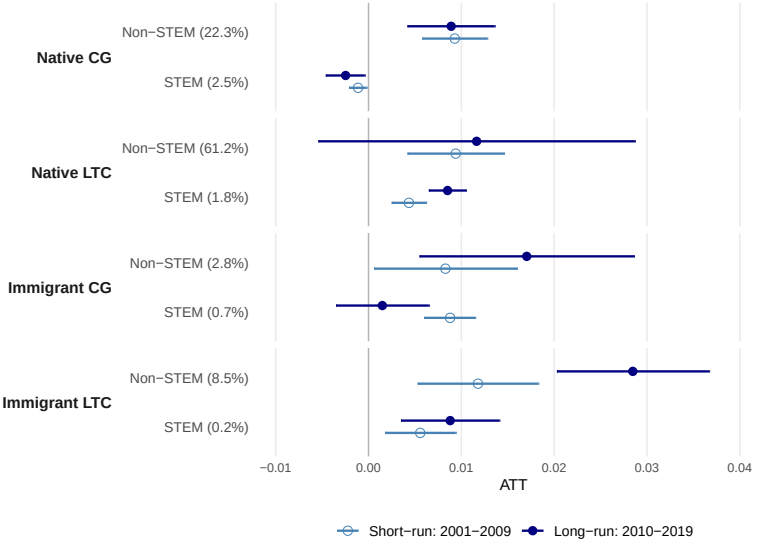
Nominal wages: σ_u=0.79, σ_l=1.06



Nominal wages: σ_u=10, σ_l=1 (substitutes)

◂ Return

H-1B: Income Effects by Group and Occupation



Abramitzky, Boustan, Gulek, Hainmueller (2026) — preliminary. Income gains concentrate in non-STEM occupations.

◂ Return

Model Setup: Labor and Intermediate Good Demand

- N firms indexed by i, R regions indexed by r, each region endowed with L_r labor

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$$\min_{\{x_{ij}\}_{j=1}^{n}, L_i} \sum_{j=1}^{n} p_j x_{ij} + w_{r_i} L_i \quad \text{subject to}$$

$$A_i(\eta_i m_i^{\frac{\sigma_u - 1}{\sigma_u}} + (1-\eta_i) L_i^{\frac{\sigma_u - 1}{\sigma_u}})^{\frac{\sigma_u}{\sigma_u - 1}} \geq y_i$$

$$m_i = \left(\sum_{j=1}^{n} \alpha_{ij} x_{ij}^{\frac{\sigma_l - 1}{\sigma_l}} \right)^{\frac{\sigma_l}{\sigma_l - 1}}$$

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$$m_i = \left(\sum_{j=1}^{n} \alpha_{ij} x_{ij}^{\frac{\sigma_l - 1}{\sigma_l}} \right)^{\frac{\sigma_l}{\sigma_l - 1}}$$

- Firms have constant and exogenous markup μ_i, and therefore set price $p_i = \mu_i C_i$

Model Setup: Labor Supply and Final Good Demand

Labor supply

- inelastically supplied in each region
- immobile across regions
- perfectly mobile across firms within region

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- perfectly mobile across firms within region

Final Demand

- All final goods consumption and firm ownership are local
- Regional Rep. consumer optimizes Cobb-Douglas utility subject to budget constraint

$$\max_{\{x_{0,i}\}} \Pi_{i \in r} x_{0,i}^{\beta_i} \quad s.t. \quad \sum_{i \in r} p_i x_{0,i} = w_r L_r + \sum_{i \in r} \pi_i$$

Notation: Input-Output Matrices

- The revenue-based input-output matrix Ω is the $(N+R) \times (N+R)$ matrix whose (ij)th element is equal to firm i's expenditure on inputs from firm j as a share of its total revenues

$$\Omega_{ij} = \frac{p_j x_{ij}}{p_i y_i}$$

- The cost-based input-output matrix $\tilde{\Omega}$: $\tilde{\Omega}_{ij} = \frac{p_j x_{ij}}{\sum_{k=1}^{N+R} p_k x_{ik}}$

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- Leontief inverse matrices are defined as:

$$\Psi = (I - \Omega)^{-1} = I + \Omega + \Omega^2 + \ldots, \quad \text{and} \quad \tilde{\Psi} = (I - \tilde{\Omega}) = I + \tilde{\Omega} + \tilde{\Omega}^2 + \ldots$$

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- Labor share of firms: $\tilde{\Omega}_{i,L} = \frac{w_r L_i}{\sum_{k=1}^{N} p_k x_{ik} + w_r L_i}$
- For intuition: take the change in prices $d\ln p$ and $d\ln w$ as given

Effects of a Labor Supply Shock on Labor Share

Proposition

In response to an immigration-induced wage shock, the following equation describes the change in the labor share of production costs

$$d \ln \tilde{\Omega}_{i,L} = (1 - \sigma_u)(1 - \tilde{\Omega}_{i,L})(d \ln w_{r_i} - \sum_{j=1}^{n} \frac{\tilde{\Omega}_{ij}}{\tilde{\Omega}_{iM}} d \ln p_j)$$

Effects of a Labor Supply Shock on Labor Income

- Revenue-based Domar weights λ_i of producer i is its sales as a fraction of nominal GDP

$$\lambda_i \equiv \frac{p_i y_i}{nGDP} = p_i y_i$$

Effects of a Labor Supply Shock on Labor Income

- Revenue-based Domar weights λ_i of producer i is its sales as a fraction of nominal GDP

$$\lambda_i \equiv \frac{p_i y_i}{nGDP} = p_i y_i$$

- The covariance operator is defined by:

$$Cov_{\tilde{\Omega}^{(j)}}(d \ln p, \Psi_{(k)}) = \left(\sum_{i=1}^{N+R} \tilde{\Omega}_{ji} d \ln p_{(i)} \Psi_{ik} \right) - \left(\sum_{i=1}^{N+R} \tilde{\Omega}_{ji} d \ln p_i \right) \left(\sum_{i=1}^{N+R} \tilde{\Omega}_{ji} \Psi_{ik} \right)$$

where

- $\tilde{\Omega}^{(j)}$ corresponds to the jth row of $\tilde{\Omega}$
- $\Psi_{(k)}$ is the kth column of Ψ
- $d \ln p$ is the vector of price changes of all inputs

Effects of a Labor Supply Shock on Sales

Proposition

In response to an immigration-induced wage shock, the following equation describes the change in the Domar weights (sales)

$$
\begin{aligned}
d \ln \lambda_i = & \sum_{j=1}^{n} (1 - \sigma_l) \frac{\lambda_j}{\lambda_i \mu_j} Cov_{\tilde{\Omega}^{(j)}} \left(d \ln p, \Psi_{(i)} \right) \\
& + (\sigma_u - \sigma_l) \sum_{j=1}^{n} \frac{\lambda_j}{\lambda_i} \tilde{\Omega}_{j,L} \left(d \ln w_{r_j} - \sum_{k=1}^{n} \frac{\tilde{\Omega}_{j,k}}{\tilde{\Omega}_{j,M}} d \ln p_k \right) (\Psi_{ji} - I_{ji}) \\
& + \textit{Pass-through}
\end{aligned}
$$

where I is the identity matrix

Change in Prices

Proposition

In response to an immigration-induced wage shock, the following equation describes the change in prices charged by firms

$$d \ln p_i = \sum_{j=1}^{n} \tilde{\Psi}_{ij} \tilde{\Omega}_{jL} d \ln w_{r_j}$$

Structural Equations

Combining propositions 1–3, introducing time t, and replacing $d \ln w_{rt} = \frac{\delta_{rt}}{\epsilon^D}$ gives:

$$d \ln \tilde{\Omega}_{it,L} = -\frac{(1-\sigma_u)}{\epsilon^D} \underbrace{\sum_{r=1}^{R} \tilde{\Psi}_{i,r} \delta_{rt}}_{\text{U: Upstream exposure}} + \eta_{it,1}$$

Structural Equations

Combining propositions 1–3, introducing time t, and replacing $d \ln w_{rt} = \frac{\delta_{rt}}{\epsilon^D}$ gives:

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$$d \ln \lambda_{it} = \underbrace{\frac{(1-\sigma_l)}{\epsilon^D} \sum_{j=1}^{n} \frac{\lambda_j}{\lambda_i \mu_j} Cov_{\tilde{\Omega}^{(j)}} \left(\sum_{r=1}^{R} \tilde{\Psi}_{(r)} \delta_{rt}, \Psi_{(i)} \right)}_{\text{D1: Downstream exposure 1}}$$

$$+ \underbrace{\frac{(\sigma_u - \sigma_l)}{\epsilon^D} \sum_{j=1}^{n} \frac{\lambda_j}{\lambda_i} \tilde{\Omega}_{j,L} \left(\delta_{r_{j,t}} - \sum_{k=1}^{n} \frac{\tilde{\Omega}_{j,k}}{\tilde{\Omega}_{j,m}} \left(\sum_{r=1}^{R} \tilde{\Psi}_{k,r} \delta_{rt} \right) \right) (\Psi_{ji} - I_{ji})}_{\text{D2: Downstream exposure 2}} + \eta_{it,2}$$

Full Characterization

Theorem

The change in equilibrium prices and quantities in response to an immigration shock is given by the solution to a system of linear equations which takes two inputs: statistics of the production network at baseline and the structural elasticity parameters

Full system Return

Full characterization: the linear system

$$
\begin{aligned}
d\ln\lambda_r =& \sum_{i\in r}\frac{\lambda_i\Omega_{iL}}{\lambda_r}(d\ln\lambda_i + d\ln\Omega_{iL})\\
d\ln\Omega_{i,L} =& (1-\sigma_u)(d\ln w_{r_i} - \sum_{j=1}^{n}\frac{\tilde{\Omega}_{ij}}{\tilde{\Omega}_{iM}}d\ln p_j)\\
d\ln\lambda_i =& (1-\sigma_l)\sum_{j=1}^{n}\frac{\lambda_j}{\lambda_i\mu_j}Cov_{\tilde{\Omega}^{(j)}}\left(d\ln p, \Psi_{(i)}\right)\\
&+ (\sigma_u-\sigma_l)\sum_{j=1}^{n}\frac{\lambda_j}{\lambda_i}\tilde{\Omega}_{j,L}\left(d\ln w_{r_j} - \sum_{k=1}^{n}\frac{\tilde{\Omega}_{j,k}}{\tilde{\Omega}_{j,M}}d\ln p_k\right)(\Psi_{ji} - I_{ji})\\
&+ \frac{1}{\lambda_i}\sum_j\sum_r b_{rj}\Psi_{ji}\chi_r d\ln\chi_r\\
d\ln\chi_r =& \left(\sum_{i\in r}\frac{\pi_i}{\chi_r}d\ln\lambda_i\right) + \frac{\lambda_r}{\chi_r}d\ln\lambda_r\\
d\ln p_i =& \sum_{j=1}^{n}\tilde{\Psi}_{ij}\tilde{\Omega}_{jL}d\ln w_{r_j}\\
d\ln w_r =& d\ln\lambda_r - d\ln L_r
\end{aligned}
$$

◂ Return

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